



< Previous

Next >

Notifications

1. Eigenvalues, eigenvectors, and diagonalization

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Objectives

• Find the **characteristic polynomial** of an $n \times n$ matrix.

• Find the **eigenvalues** of an $n \times n$ matrix and for each eigenvalue the **eigenvectors /eigenspace** .

• **Diagonalize** a **complete (not deficient)** $n \times n$ matrix.

• Know the relationship between eigenvalues and the **trace and determinant** of an $n \times n$ matrix.

• Understand how to apply the theory of eigenvectors and eigenvalues to the **pagerank** algorithm used by Google and other search engines.

1. Eigenvalues, eigenvectors, and diagonalization

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1/2

1/14/24, 10:55 PM

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